A LOCALLY FINITE TYPE MORPHISM OVER A LOCALLY NOETHERIAN BASE HAS LOCALLY NOETHERIAN SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a direct presentation reduction; every step names the definition or the scheme statement it uses.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The presentation of a morphism of algebraic spaces by a square of schemes. used at proof lines 4688-4697. Supplies U, V and the morphism U -> V.
- Local finite typeness and local Noetherianness are read on a presentation. used at proof lines 4697-4699. Translate the hypotheses on f and Y into hypotheses on U -> V and V.
- The scheme case: locally finite type over a locally Noetherian base is locally Noetherian. Morphisms, Lemma morphisms-lemma-finite-type-noetherian, used at proof lines 4699-4701. Gives that U is locally Noetherian.
- Local Noetherianness of an algebraic space defined through a presentation. used at proof lines 4701-4702. Concludes for X.

Source reference. Original lemma 01T6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2363–2363. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally of finite type and Y is locally Noetherian, then X is locally Noetherian.*

Proof. Let

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

be a commutative diagram where U, V are schemes and the vertical arrows are surjective étale. If f is locally of finite type, then $U \to V$ is locally of finite type. If Y is locally Noetherian, then V is locally Noetherian. By Morphisms, Lemma 01T6 we see that U is locally Noetherian, which means that X is locally Noetherian. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.